Lab 4

Launch SCORM

QUESTION 3

From the code, what is P(maintenance = yes | rain = light)?

0.2 Check Hint

Correct!

0.2. If the rain is light, there is a 20% chance of maintenance.

QUESTION 4

From the code, what is P(train = delayed | rain = heavy, maintenance = no)?

0.5 Check Hint

Correct!

0.5. With heavy rain and no maintenance, the train is late half the time.

QUESTION 5

How many numbers are in the Train table in total?

12 Check Hint

Correct!

2 rows × 6 columns = 12. More parents means bigger tables, very fast.

QUESTION 13

Your hand answer and the code are the same: 0.4601. Why?

It is luck

The code guesses with random samples

The code does exact inference: the same sums and normalization, in a faster order

Correct!

VariableElimination is an ex Launch SCORM computes the same sums as enumeration but reuses partial results, so it stays fast on bigger networks.

Variable elimination in pgmpy computes exact conditional probabilities by performing the same underlying joint summations and normalization as hand calculation.

Hand sum = 0.098 + 0.064 + 0.051 = 0.213
Normalized value = 0.098/0.213 = 0.4601

QUESTION 14

Why exactly 0.4?

Appointment depends only on Train, and Train is known, so we read it straight from the table

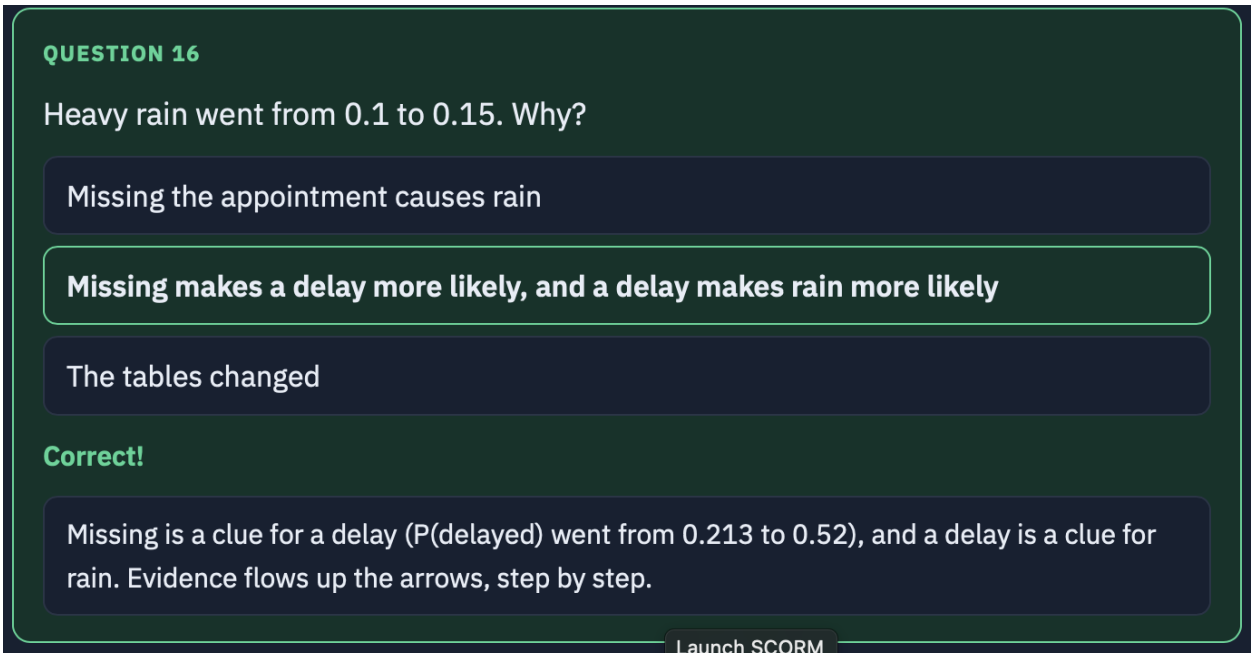
It is a coincidence

Because the rain is heavy

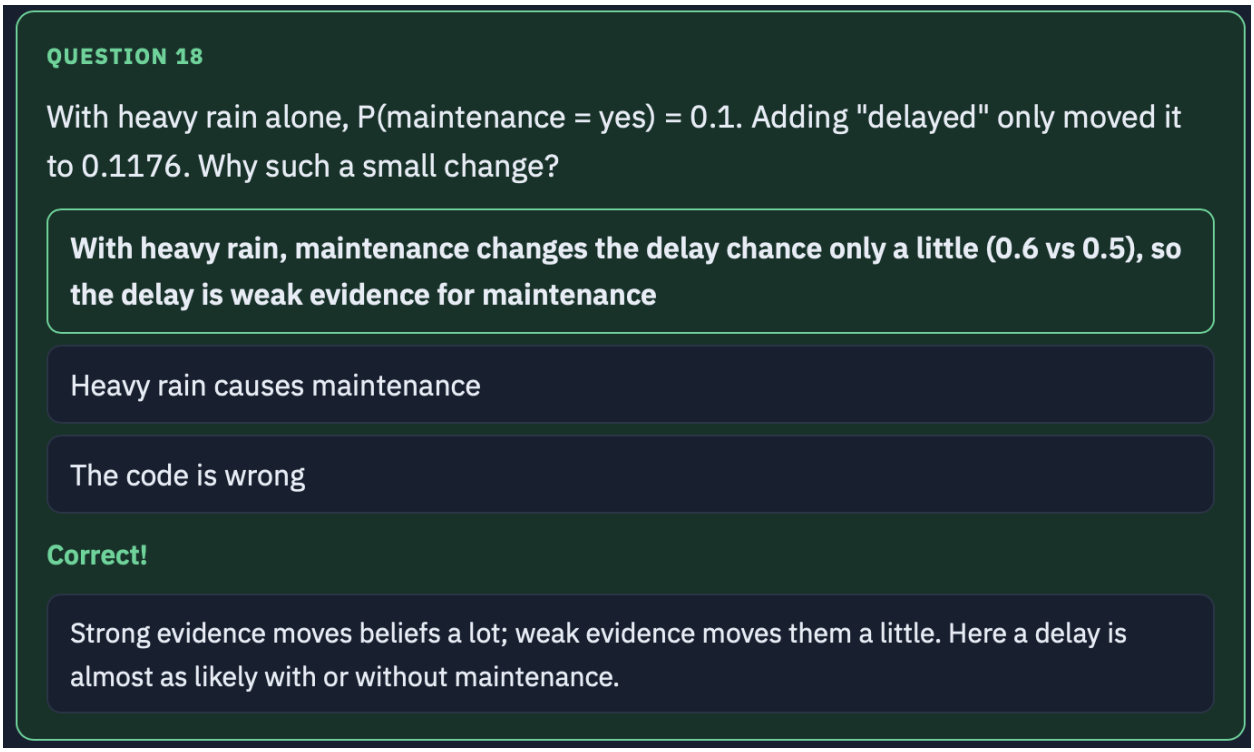
Correct!

Train is Appointment's only parent. Once we know the parent, the nodes above it (Rain, Maintenance) cannot change anything for Appointment.

Appointment is conditionally independent of all other nodes given its parent Train. Therefore, P(appointment = miss |train = delayed) is read directly from its CPT entry (0.4).



P(rain = heavy | appointment miss) = 0.0253/0.1639 = 0.15



P(m = yes | heavy, delayed) = 0.1 x 0.6/(0.1 x 0.6) + (0.9 x 0.5) = 0.06 / 0.06 + 0.45 = 0.06 / 0.51 = 0.1176